

BINLEVEL · CALCULATION

BinLevel sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.3	2026-09-30	Amish Chadha	Design for construction (BNL-DDR-003); 55 mm box, no tabs, studs and standoffs, M6 x 20 bolts; geometry, mass, size, fixing and cost re-run; new check J3
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); 2 mm aluminium bracket, R6 to 70 °C, gated rate-of-rise rule, 2 h interval at SF12 only
0.1	2026-09-25	Amish Chadha	First issue for TRL 3 (measurement geometry, accuracy, energy, supply, airtime and payload, link budget, timing, lid temperature, mass, shock, cost)

BinLevel sizing calculations

On paper, BinLevel meets ten of its sixteen requirements (six by calculation and four by design), has two at risk, misses two and leaves two that only a test can settle. The two misses are R5 (the stock enclosure is IP67, not IP69K) and R10 in steel containers (an antenna inside a steel container loses the link beyond about 0.35 km; the choice of remedy is still open). The two at risk are accuracy on real waste (R2) and the climate range (R6: a dark lid reaches about 67 °C against the new 70 °C limit, and the transducer's rating is unconfirmed). Version 0.2 applies the recommendations Amish accepted on 2026-09-25 (BNL-DDR-002): a 2 mm aluminium bracket brings the unit from about 435 g to 334 g, so R16 is now met; R6's upper limit rises from 60 to 70 °C; the 15 K rate-of-rise heat trigger counts only above 50 °C, which removes the false alarm when sun breaks through cloud, so R9 is now met on paper; and the routine interval stretches to 2 h at SF12 only. Version 0.3 re-runs the geometry, mass, size, fixing and cost checks for the constructable design of BNL-DDR-003 (a 55 mm tall box held on four studs and standoffs, no tabs, shorter bolts and the parts added for construction): the unit is now about 350 g and 71.5 mm below the lid, still inside R16, parts cost \$59.00 against \$60, and no requirement changes status. Battery life is not a constraint: a C cell lasts about 16 years in the worst radio case. The calculations changed four details of the TRL 2 concept: the payload shrinks from 12 to 11 bytes so that it fits the slowest US915 data rate, the carrier gains a nanopower regulator because a fresh cell exceeds the module's 3.6 V limit, the temperature is read every 5 min so that the heat alert arrives within 15 min, and the cell is strapped rather than held by clips alone. Every number in this note is printed by `docs/04-calcs/sizing.py`; the tag in brackets, for example [C3], is the line of that script's output that carries it.

Safety: These are first-principles estimates for a paper proof of concept. The temperature alert is a maintenance aid and not fire detection. Nothing here replaces checks of the lithium cell's fusing and retention, or safe working practice around bins and collection vehicles. See BNL-PRC-001, Safety.

■ Scope and method

The note checks every requirement in BNL-REQ-001 v0.5 against the design in BNL-PRC-001 v0.5 and the parametric model `cad/src/model.py`. The script imports the model's `PARAMS`, `derived()` and part solids, so the container depth, sensor positions, bracket volume and envelope used here are the ones in the STEP files and in drawing BNL-DWG-001. The script also reads `bom/bom.csv` and `budget_usd` in `project.yaml`. Run it from the repo root with `python docs/04-calcs/sizing.py`.

The design case is an 1,100 L four-wheel communal container (DDR-001, D1) with the sensor at the center of an HDPE lid, ranging every 15 min, reporting hourly (2 h at SF12 only, DDR-002) and on alerts (D6), on a LoRaWAN network with a TwinKit or public gateway (D4, D5).

■ Assumptions

Table 1. Main assumptions.

Area	Assumption	Basis
Container	Body 1,300 x 1,000 mm outside, 8 mm wall, inner floor 228 mm and rim 1,300 mm above ground; lid wall 5 mm HDPE	Typical EN 840 1,100 L size (estimate), as in the model
Sensing	Ultrasonic blind zone 0.25 m and range to 4 m; effective beam half-angle 15°; ToF used to 0.4 m with a 27° field of view and ± 20 mm error including the window	JSN-SR04T-class figures are estimates to confirm; the VL53L1X field of view and 4 m maximum are from the ST product page
Accuracy	Air temperature known to ± 5 K from the on-board sensor; echo detection jitter of half a wavelength; 1 μ s timer	Assumed
Module	RAK3172: 2.0 to 3.6 V supply, 1.69 μ A sleep, -20 to 85 °C (the -T variant -40 to 85 °C)	RAK3172 datasheet
Currents	Transmit 45 mA at +14 dBm (the datasheet gives only 87 mA at +20 dBm); receive 6 mA for at least 0.1 s or 8 symbols per window; 0.5 s of processing at 3 mA per uplink; sleep 6.0 μ A in all (Section C)	Assumed, to be measured
Ranging cycle	MCU 3 mA for 0.5 s; ultrasonic module 10 mA for 0.4 s (power-up and five pings 60 ms apart); ToF 20 mA for 0.1 s	Assumed typical module figures
Traffic	Two extra uplinks a day for alerts and tip events	Assumed
Cell	Li-SOCl ₂ C (ER26500 class) 8.5 Ah or AA (ER14505 class) 2.6 Ah nominal; 60 % usable for pulses, cold and end-of-life voltage; self-discharge 1 % of nominal a year; 3.67 V open circuit	Typical bobbin-cell data
Radio	11-byte payload plus 13 bytes of LoRaWAN overhead; 125 kHz; coding rate 4/5; 8-symbol preamble; low data rate optimization at SF11 and SF12; US915 slowest rate (DR0, SF10) carries at most 11 bytes of application payload and a 400 ms dwell limit (TTN for the dwell limit; the 11-byte figure is from the LoRaWAN Regional Parameters and was not re-fetched this session)	LoRaWAN defaults, as in FND-CAL-001
Link	+14 dBm; 0 dBi node antenna with 0.5 dB loss; 2 dBi gateway antenna with 2 dB feeder loss, 30 m high; node 1.3 m high; 6 dB noise figure; Okumura-Hata urban model at 868 MHz; container penalty 10 dB (HDPE, antenna under the lid), 30 dB (steel, antenna inside), 3 dB (steel, external lid antenna); 10 dB fade margin	Screening values; container penalties are assumptions to be measured
Lid heat	40 °C ambient; 900 W/m ² sun (100 W/m ² under cloud); absorptance 0.9 (dark lid) or 0.5 (light lid); 25 W/m ² K above and 5 W/m ² K below the lid; unit heat capacity 400 J/K coupled to the lid at 0.75 W/K	Handbook ranges; assumed coupling
Shock	20 g peak when the container strikes the lifter stop; HDPE shear strength 20 MPa; 18 mm washers; M6 stress area 20.1 mm ² at 210 MPa	Assumed; the 20 g figure needs measurement
Mass	ABS 1.05, stainless 7.9, aluminium 2.68 and FR-4 1.85 g/cm ³ applied to model volumes, with ASA 1.07 and PMMA 1.19 g/cm ³ for the printed mounts and window; bracket 2.0 mm 5052-class aluminium (DDR-002), no tabs (DDR-003); bought-in parts 174 g (Section I)	Assumed part masses

■ A. Measurement geometry (R1)

- **Depth.** The lid underside is 1,072 mm above the inner floor, and the transducer face sits 1,000 mm above it (1,011 mm before the box grew 10 mm taller under BNL-DDR-003) [A1].
- **Blind zone.** At the adopted 80 % threshold (D6) the waste surface is 200 mm below the face, inside the 250 mm ultrasonic blind zone, so the ultrasonic sensor alone reads fill only up to 75.0 % [A2]. This confirms decision D2.
- **Coverage.** The ToF sensor, used to 400 mm, covers fill from 60.0 % to 100 %, which leaves a 150 mm overlap band where both sensors read and can be compared [A3]. Together they cover the 0 to 1.00 m needed, inside the R1 target of 0.03 to 1.5 m [A4].
- **Beam.** A 15° beam spreads to 0.54 m across at the floor, which averages over uneven waste; the ToF footprint at 0.4 m is 192 mm across [A5]. Mounted at the lid center, the transducer is 492 mm from the nearest wall, so the beam clears the walls down to the floor for half-angles up to 26.2° [A6]. An off-center mount would see echoes from the walls sooner.

■ B. Instrument accuracy (R2)

- **Speed of sound.** Over the revised R6 range of -20 to 70 °C the speed of sound runs from 318.9 to 371.3 m/s; without compensation the distance error would be -7.1 % to +8.2 % [B1]. The on-board temperature reading removes most of it.
- **Error budget.** With a 5 K error between the sensor and the air column, the residual is 0.85 %, or 8.5 mm at full depth; echo jitter adds 4.3 mm and the timer 0.17 mm [B2]. The ultrasonic instrument error is 9.5 mm and the ToF error 20 mm, against a ± 100 mm allowance (± 10 % of depth), so the instruments use 10 % and 20 % of it [B3].
- **Conclusion.** R2 is decided by the waste surface (bags, cardboard, voids and slopes), not by the sensors. No calculation settles it, so R2 stays at risk.

■ C. Energy and battery life (R4)

Table 2. Sleep current [C1].

Item	Current
Module in stop mode (datasheet 1.69 μ A, rounded up)	2.0 μ A
Accelerometer in wake-on-motion mode (LIS2DH12 class)	2.0 μ A
Temperature sensor in shutdown	0.5 μ A
Nanopower regulator, load switch and divider leakage	0.5 μ A
Buffer capacitor leakage	1.0 μ A
Total	6.0 μA

- **Daily use.** Sleep takes 0.144 mAh a day; each ranging cycle takes 7.50 mAs, or 0.200 mAh a day at 96 cycles; temperature reads every 5 min add 0.0012 mAh a day [C1]. An uplink takes 3.32 μ Ah at SF9 and 19.82 μ Ah at SF12; with two alerts a day the uplinks use 0.086 mAh (SF9, hourly) or 0.278 mAh (SF12, 2-hourly) a day [C2].

Table 3. Daily charge and life on 60 % of nominal capacity [C3].

Case	C cell (8.5 Ah)	AA cell (2.6 Ah)
SF9, hourly	0.664 mAh/day; 21.0 years	0.503 mAh/day; 8.5 years
SF12, every 2 h	0.856 mAh/day; 16.3 years	0.694 mAh/day; 6.2 years

- **R4 is met.** The worst case averages 35.7 μ A including self-discharge and gives 16.3 years on a C cell and 6.2 years on an AA cell, against 5 years minimum and a 10-year design target [C4]. The TRL 2 estimates (about 23 and 18 years) were optimistic by the extra sleep current and the alert traffic, but the conclusion stands: seals and plastics, not the cell, set the service life. The AA cell meets the 5-year minimum but not the 10-year target, which supports decision D3.

■ D. Supply voltage, pulses and heating (R6, R4)

- **Regulator.** A fresh Li-SOCl₂ cell rests at 3.67 V, 70 mV above the RAK3172's 3.6 V maximum [D1]. The carrier therefore needs a nanopower 3.3 V regulator (quiescent current well under 1 μ A, included in Table 2). This is a change from TRL 2, added to BOM item 6 at no change in its indicative price.
- **Pulse load.** Buffering a whole SF12 uplink (45 mA for 1.48 s) in a capacitor with 0.5 V droop would need 0.133 F [D2], and a supercapacitor of that size leaks several microamperes. The design lets the cell carry the pulse, specifies a cell rated for at least 100 mA pulses, and adds a 1,000 μ F low-leakage buffer to cover the voltage delay of a passivated cell. Cold-start voltage delay at -20 °C remains to be tested.
- **Heating is not an option.** A 20 mW heater to keep the ToF window above the dew point would use 133 mAh a day, 156 times the whole budget [D3]. Condensation must be handled by venting, a hydrophobic window and the ultrasonic sensor's tolerance to it.

■ E. Airtime, payload and radio rules (R10, R3)

Table 4. Time on air for an 11-byte uplink (24 bytes on air) at 125 kHz, with two alerts a day [E1].

SF	Per uplink	Hourly (s/day)	Every 2 h (s/day)	EU868 1 % off-time
7	62 ms	1.6	0.9	6 s
8	113 ms	2.9	1.6	11 s
9	206 ms	5.4	2.9	20 s
10	371 ms	9.6	5.2	37 s
11	823 ms	21.4	11.5	82 s
12	1,483 ms	38.6	20.8	147 s

- **Payload size.** The TRL 2 concept used 12 bytes. The slowest US915 rate (DR0, SF10) carries at most 11 bytes of application payload, so a 12-byte payload could not be sent from a US915 sensor at the edge of coverage. The payload is cut to 11 bytes, which fits [E2]; the airtimes are unchanged from TRL 2 because the symbol count does not change. At SF10 an uplink takes 371 ms, inside the 400 ms US915 dwell limit [E3].
- **Fair use.** Hourly reporting at SF12 would use 38.6 s a day, over TTN's 30 s, while SF11 hourly uses only 21.4 s. The routine interval therefore stays at 1 h up to SF11 and stretches to 2 h at SF12 only (DDR-002; D6 had

stretched it at SF11 as well), which keeps SF12 at 20.8 s [E4]. Battery life is unchanged, because SF12 at 2 h remains the worst case in Section C.

- **Payload layout.** A candidate 11-byte layout, for the decoder shared with FieldNode (D5): 1 byte version and flags (alert, tip, lid, low battery, sensor used), 1 byte fill in 0.5 % steps, 2 bytes distance in millimeters, 1 byte temperature in degrees Celsius (signed), 1 byte lid-open count, 1 byte tip count, 1 byte battery voltage in 10 mV steps above 2.0 V, 2 bytes minutes since the last emptying and 1 byte spare. This is a sketch for review, not firmware.

■ F. Link budget (R10)

Table 5. Link margin at 1 km and range with a 10 dB fade margin [F1, F2].

Case	SF9 (143.0 dB budget)	SF12 (150.5 dB budget)
HDPE container, antenna under the lid (10 dB)	6.5 dB; 0.80 km	14.0 dB; 1.30 km
Steel container, antenna inside (30 dB)	-13.5 dB; 0.22 km	-6.0 dB; 0.35 km
Steel container, external lid antenna (3 dB)	13.5 dB; 1.26 km	21.0 dB; 2.06 km

- Urban path loss at 1 km is 126.5 dB; gateway sensitivity is -129.5 dBm at SF9 and -137.0 dBm at SF12 [F1].
- **Plastic containers** work to about 0.8 km at SF9 and 1.3 km at SF12 with a 10 dB fade margin. A pilot should place gateways within about 1 km of the containers, or accept SF12 and the 2 h interval.
- **Steel containers with the internal antenna do not meet R10:** even at SF12 the range is only about 0.35 km. An external lid antenna restores 1.3 to 2.1 km (open item O1 in DDR-001).

■ G. Timing (R3, R9)

- **Fill alert.** In the worst case the surface crosses 80 % just after a reading: 15 min to the next reading, up to 147 s of EU868 off-time after a previous SF12 uplink, and 1.5 s of airtime give 17.5 min, inside R3's 20 min [G1].
- **Heat alert.** Read only with the 15 min ranging, a heat alert could take 17.5 min, which misses R9's 15 min. Reading the temperature every 5 min (0.0012 mAh a day [C1]) brings it to 7.5 min [G2]. This is a change from TRL 2.

■ H. Temperature under the lid (R6, R9)

- **Hot lids.** In 900 W/m² sun at 40 °C ambient a dark lid reaches about 67 °C and a light lid about 55 °C [H1]. The unit is bolted to the lid underside and runs close to lid temperature, so on dark lids in hot climates it exceeded R6's former 60 °C limit. R6's upper limit is now 70 °C (DDR-002), which covers this with 3 K of margin. The module (to 85 °C) and cell tolerate it; the ultrasonic module's rating must be confirmed, so R6 stays at risk.
- **Cold.** The standard RAK3172 is rated to -20 °C, exactly R6's lower limit; the RAK3172-T (-40 to 85 °C) is named for cold sites (DDR-002).
- **False heat alarms.** The lid's time constant is 5.0 min and the unit's 8.9 min. When 800 W/m² sun breaks through cloud, a dark lid rises 24 K and the unit 15.4 K within 15 min [H2], just over the 15 K rate-of-rise trigger (D7). With the original rule the rate trigger could fire on a sunny afternoon with no fire.
- **Gated rate rule (DDR-002).** The 15 K rise now counts only when the unit is already above 50 °C; the 70 °C absolute alert is kept. Under cloud the unit sits at about 43 °C, below the gate, so the sun step does not trigger,

and the steady dark lid (about 67 °C) stays below 70 °C [H3]. In the worst case above the gate, a unit already at 50 °C that then sees full sun rises 10.9 K in 15 min, below the 15 K trigger [H4]. R9 is met on paper; the thermal model is a screening estimate and must be checked by measurement. The alert remains a maintenance aid, not fire detection.

■ I. Mass and size (R16)

- **Mass.** Enclosure 94 g (base and cover, 55 mm tall), bracket plate 63 g (2.0 mm 5052-class aluminium, no tabs), electronics board 13 g, printed collar and ToF holder with the window 5 g, and bought-in parts 174 g (now including the box fixings, 22 g, and the shorter bolts, 36 g) give 350 g, inside R16's 400 g [I1]. **R16 is met.** Version 0.2 gave 334 g with the 45 mm box and tabs; version 0.1 used a 1.5 mm stainless bracket (unit 435 g). A 2.0 mm stainless plate would make the unit 473 g [I2].
- **Size.** Below the lid the unit measures 150 x 80 x 71.5 mm (61.5 mm before the box grew 10 mm taller), inside the 160 x 90 x 100 mm limit [I3].
- **Corrosion.** The bolts stay stainless. Stainless bolts through wet aluminium can corrode the aluminium around the holes; an anodized plate or insulating washers should be considered (review suggestion, not applied).

■ J. Shock and fixing (R7)

- A 20 g shock on the 0.35 kg unit gives 69 N. One washer pulling through the 5 mm HDPE lid needs about 5,655 N and one M6 bolt carries about 4,221 N [J1], so the fixing has a large margin on strength. Loosening under repeated shocks, and IK08 impact, need a test.
- The 50 g cell needs 10 N of retention at 20 g [J2]; the holder is specified with a strap rather than spring clips alone.
- **Box fixing (new in v0.3).** The box and everything in it, about 250 g, hang on the four M4 studs: 49 N at 20 g, or 12 N per stud. The base floor would need about 2,100 N per stud to pull through under a 9 mm sealing washer (ABS shear strength 30 MPa, assumed), and a press-in stud's push-out rating in 2 mm aluminium is taken as about 900 N (to confirm from the maker's data) [J3]. Strength is not the limit; loosening needs a test.

■ K. Cost (R14)

- All eleven BOM lines are priced; the parts total is \$59.00 against the \$60 value-engineering target (`budget_usd`), \$1.00 under [K1]. Lines 10 (box fixings) and 11 (printed mounts and window) were added for construction (BNL-DDR-003). The design decisions register lists the cost drivers and savings worth trying.
- An external lid antenna for steel containers (about \$8, open item O1) would take the total to \$67.00, \$7.00 over the value-engineering target [K2].

■ L. Results against every requirement

Table 6. Requirement status at TRL 3.

ID	Requirement	Target	Value (tag)	Status
R1	Range	0.03 to 1.5 m below the face	0 to 1.00 m needed; ultrasonic 0.25 to 4 m, ToF to 0.4 m (A2 to A4)	Met (datasheet ranges to confirm)
R2	Accuracy on real waste	±10 % of depth for 90 % of readings	Instrument 9.5 mm (ultrasonic), 20 mm (ToF) of 100 mm; surface unknown (B3)	At risk
R3	Reporting and alert timing	Uplink at least every 2 h; alert within 20 min	1 h (2 h at SF12); 17.5 min (G1)	Met
R4	Battery life	5 years at SF12; 10-year target	16.3 years (C cell) (C4)	Met
R5	Sealing	IP67 minimum; IP69K where pressure washed	Stock IP67 enclosure	Not met (IP69K)
R6	Climate range	-20 to +70 °C inside the container (was +60 °C)	Dark lid about 67 °C (H1); standard module at its -20 °C limit, RAK3172-T for cold sites; transducer rating unconfirmed	At risk
R7	Lifting, tipping and lid slams	No loosening; IK08	69 N against 4,221 N per bolt (J1); 12 N per stud (J3)	Not verifiable at TRL 3
R8	Emptying and lid events	Tip event with the next uplink	Accelerometer wake-on-motion	Met by design
R9	Heat warning	Alert within 15 min above 70 °C, or on a 15 K rise in 15 min counted only above 50 °C	7.5 min with 5 min reads (G2); sun step starts at about 43 °C, below the gate (H3); 10.9 K above the gate (H4)	Met on paper (to be measured)
R10	Open LoRaWAN within radio rules	Class A; EU868, US915, IN865; duty cycle and 30 s/day	20.8 s/day at SF12 (E4); 11 bytes fits US915 DR0 (E2); steel container 0.35 km (F2)	Not met (steel containers); met for HDPE
R11	Privacy	No camera or microphone; limited payload	BOM and 11-byte layout (E)	Met by design
R12	Installation	10 min or less, four bolts, no cables	Four through-bolts on a 130 x 60 mm pattern	Not verifiable at TRL 3
R13	Tamper resistance	Inside the lid; tamper-resistant fasteners	Button-head tamper bolts from above	Met by design
R14	Cost	\$60 value-engineering target	\$59.00 (K1)	Met, within the target
R15	Open and interoperable	Documented payload; open decoder	Candidate layout in Section E; decoder shared with FieldNode (D5)	Met by design
R16	Mass and size	400 g or less; within 160 x 90 x 100 mm below the lid	350 g (I1); 150 x 80 x 71.5 mm (I3)	Met

Summary: met 10 (R1, R3, R4, R9, R14, R16 by calculation; R8, R11, R13, R15 by design); at risk 2 (R2, R6); not met 2 (R5, R10 in steel containers); not verifiable at TRL 3, 2 (R7, R12). Version 0.1 had met 8, at risk 3 and not met 3.

■ Checks against the TRL 2 figures

TRL 2 claim	TRL 3 value	Change made
Daily use about 0.61 to 0.76 mAh; life about 23 to 18 years (C)	0.664 to 0.856 mAh; 21.0 to 16.3 years	Precis and README updated
AA cell about 10 to 7 years	8.5 to 6.2 years	Precis updated
12-byte payload	11 bytes, so it fits US915 DR0	Precis, requirements and media updated
Uplink airtime about 5 s/day (SF9) and 18 s/day (SF12)	5.4 and 20.8 s/day with two alerts	Precis updated
Mass about 250 g; R16 met	435 g with stainless (v0.1); 334 g with the aluminium bracket (v0.2); 350 g constructable (v0.3); R16 met	Requirements, precis and README updated
150 x 80 x 63 mm below the lid	150 x 80 x 61.5 mm (v0.2); 71.5 mm with the 55 mm box (v0.3)	Requirements and precis updated
Temperature inside a dark lid up to about 60 °C	About 67 °C on a 40 °C day	Problem statement and requirements note updated
80 % fill about 0.21 m below the lid	0.20 m below the transducer face	Precis updated
Parts cost about \$55	\$55.00 (v0.2); \$59.00 with the parts added for construction (v0.3)	BOM, precis and README updated